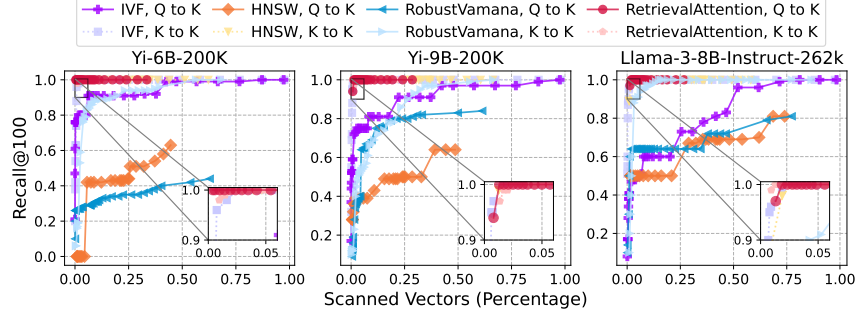


Table 5: Decoding latency breakdown on Llama-3-8B.

Methods	Retrieval	Attention	Others	Total
Flat	0.798	0.083	0.041	0.922
IVF	0.250	0.084	0.039	0.373
RetrievalAttention	0.064	0.081	0.043	0.188

Figure 6: Recall vs. scanning key vectors when using the query vector (Q to K) and key vector (K to K) as the query, individually. Q and K are dumped from three long-context LLM models.

attention vectors. The efficiency of RetrievalAttention arises because it effectively mitigates the OOD issue between query and key vectors. In contrast, for the in-distribution K to K search, all indexes exhibit good performance.

5 RELATED WORKS

To accelerate the long-context LLM inference, some works (Zhang et al., 2023; Liu et al., 2024b; Xiao et al., 2024b; Han et al., 2024; Ge et al., 2024; Li et al., 2024) attempt to compress the size of the KV cache by leveraging the sparsity of attention. However, these methods often suffer from significant model accuracy drops due to the dynamic nature of attention sparsity.

FlexGen (Sheng et al., 2023) and Lamina (Chen et al., 2024b) offload the KV cache to CPU memory, but they struggle with slow and costly full-attention computation. By identifying the dynamic nature of important KV vectors for different queries, recent work chooses to retain all of the KV cache and dynamically attend to different parts of KV vectors based on the current query. Quest (Tang et al., 2024) partitions the KV cache into blocks and selects a representative key vector for each block. For a given query, it scans all representative key vectors and attends top- k blocks with the highest attention scores. InfLLM (Xiao et al., 2024a) adopts a similar strategy as Quest but offloads most KV cache blocks to the CPU memory to support longer contexts. Due to block-based organization and retrieval, the accuracy of representative vectors significantly impacts the effectiveness of those methods for obtaining important tokens. SparQ (Ribar et al., 2024), InfiniGen (Lee et al., 2024), and LoKi (Singhanian et al., 2024) approximate the most relevant top- k keys corresponding to a given query by reducing the channel dimension. RetrievalAttention instead organizes the KV cache using ANNS indexes, allowing the retrieval of important tokens with high recalls and low cost. The concurrent work MagicPiG (Chen, 2024) and PQCache (Zhang et al., 2024a) employ LSH and PQ centroids to retrieve critical tokens, respectively. However, they fail to address the OOD issue in attention, necessitating retrieving a large portion of KV cache (e.g., 20%) for high model accuracy.

6 CONCLUSION

We propose RetrievalAttention, a method that offloads most KV vectors to CPU memory and leverages vector search for dynamic sparse attention to minimize inference cost. RetrievalAttention identifies the different distributions of the query and key vectors and employs an attention-aware approach to efficiently find critical tokens for model generation. Experimental results demonstrate that RetrievalAttention effectively achieves $4.9\times$ and $1.98\times$ decoding speedup than exact KNN and

traditional ANNS methods, on a single RTX4090 GPU for a context of 128K tokens. RetrievalAttention is the first system that supports running 8B-level LLMs with 128K tokens on a single RTX4090 (24GB) GPU with an acceptable latency cost and without compromising model accuracy.

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